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Editor-in-Chief
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Korean Society of Mechanical Engineers

Re: Submission of “Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Decomposition of Size-and-Surface Factor Sensitivity Under a Common ISO Stress Baseline”

Dear Editor,

I am pleased to submit the enclosed manuscript for consideration in the Journal of Mechanical Science and Technology.

1. Clear Problem Statement

When three engineers—using different standards (ISO, AGMA, FKM), different mean-stress corrections, and different surface-finish assumptions—compute the bending fatigue life of the *same* spur gear under the *same* load, their predictions span three orders of magnitude. Prior gear fatigue studies have compared individual standards or correction methods in isolation (e.g., Bonaiti 2024 for ISO vs. AGMA; Dowling 2009 for mean-stress corrections). The present study extends this tradition by simultaneously decomposing the contributions of five methodological choices and ranking them by variance explained.

2. Critical Methodology

A censored-likelihood AFT survival model is applied to a 144-combination full-factorial design (5 factors $\times$ 2–4 levels, 92 run-outs). Unlike ordinary ANOVA/Sobol’—which must discard all 92 run-out entries and produce systematically distorted variance fractions—the AFT framework retains right-censored observations through the EM algorithm. Bootstrap (2000 draws), Sobol’ sensitivity analysis, Weibull-AFT model selection ($\Delta\text{AIC} = +29.7$ in favor of log-normal), and full residual diagnostics (Shapiro-Wilk $p = 0.599$, Levene’s test) confirm the model specification. Sensitivity analyses for the Y_{Sa} stress concentration assumption ($\leq +2.1$ pp bias) and the application factor K_A (1.0–1.5, rankings stable) validate the robustness of the findings.

3. Consequential Findings

Three factors account for 97.9% of total log-variance: mean-stress correction (39.1%), size-and-surface standard (35.3%), and surface roughness (23.5%). The S–N curve source (−0.7%) and cumulative damage rule (0.3%, constant-amplitude only) are negligible. This ranking is stable under bootstrap resampling, load-factor variation, and alternative stress-concentration treatments. Engineering design recommendations—including calibration factors between standards and risk-classification criteria—are derived directly from the statistical results.

4. Fit to JMST Scope

The manuscript aligns with JMST's Solid Mechanics & Design section. It addresses a practical mechanical engineering problem (gear fatigue life prediction), employs rigorous statistical methods (censored AFT), quantifies engineering uncertainty sources in terms accessible to design engineers, and provides a reusable computational framework applicable to other machine elements. No physical experiments are required—the entire analysis uses publicly available S–N data and freely verifiable standard formulas—making the work fully reproducible.

5. Manuscript Details

- Manuscript type: Original Article
- Figures: 7; Tables: 3
- Supplementary material: Full reproducibility package (Python code, sweep data, AFT decomposition results, sensitivity analysis scripts)
- The manuscript has been formatted for double-blind review per JMST guidelines

I confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. I have approved the manuscript and agree with its submission to JMST.

Thank you for considering this submission. I look forward to the reviewers' feedback.

Sincerely,

Zhilei Chen